INTERNATIONAL QUANTITATIVE INVESTING CHALLENGE 2026

JUNEJA STREET

Strategy Declaration

ML Supertrend: Adaptive Reversal Strategy

Young Investors Cup 2026 gold medallists

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Investment hypothesis and complete strategy

We target temporary selling-pressure dislocations in volatile stocks, buying after a fresh low and an upward reversal. The hypothesis fails when a decline reflects lasting information or trading costs absorb the rebound. Each completed 30-minute candle updates price, volume, ATR (average true range) and RSI (relative strength index). In the background, the model tests hypothetical long and short trades over five candles. Completed probes train the machine-learning models through rolling statistics, weighted batches and memory of similar conditions. The models adjust band width and smoothing length in bounded steps. A fresh low, upward trend flip, RSI(14) below 30 within 50 candles and a ten-candle gap since the last signal produce a BUY. We process earlier signals first, with alphabetical ticker order for ties. Position size is the smaller of the 10% value cap and the 0.5% planned-loss cap, using a stop two ATR(24) below entry. Liquidity, sector and remaining portfolio limits can reduce the order further. We hold at most six long positions, with 40% total stock exposure and 2% aggregate planned stop risk. We enter after confirmation and close on SELL, a protective stop or an account loss trigger.

A probe records what a hypothetical trade would have earned from the candle open to the close of its fifth candle. A rise of one entry ATR gives the long probe a score of +1 and the short probe -1. The model uses the result only after the fifth candle closes. Probes overlap, so their results are dependent.

Batch weights equal confidence / (1 + adverse movement in ATR units): one ATR of adverse movement halves the weight at equal confidence. Context memory contributes up to 65% of the context/global blend. The model smooths the resulting target; the worked update appears below. Learning changes signal sensitivity; position sizes follow the separate loss budget.

Why these stocks

DELL is the paper-trading example. For risk calibration, DELL and MSFT supply technology exposure; JPM, UNH, GE and XOM add finance, healthcare, industrials and energy. We chose this sample to compare five sectors. The trading universe remains broader: each stock must pass its BUY, liquidity and risk checks before receiving capital.

1 Strategy and quantitative method

Use completed regular-session 30-minute candles for supported S&P 500 stocks. BUY opens one long position; repeated BUYs do not add size. SELL or an earlier protective stop closes it. Orders follow confirmation; historical replay fills at the next bar open plus costs.

1. Measure normal price movement

True Range Equation

$$TR_t = \max\{H_t - L_t,\ |H_t - C_{t-1}|,\ |L_t - C_{t-1}|\}$$

t is the current completed candle; t - 1 is the previous candle. H, L and C are high, low and close in dollars. TR takes the largest candle range or gap from the previous close. Vertical bars mean absolute value.

Volatility Envelope Equations

$$x_t = \frac{H_t + L_t + 2C_t}{4}, \qquad A_t = A_{t-1} + \frac{TR_t - A_{t-1}}{\ell_t}$$

$$b_t^- = x_t - m_t A_t, \qquad b_t^+ = x_t + m_t A_t$$

x is the close-weighted price, A the smoothed true range, l the smoothing length and m the band multiplier. The raw boundaries are b minus and b plus. Initially l = 24 and m = 1.4, so each raw boundary lies 1.4 times A from x. Doubling A doubles that distance. The lower boundary can only rise while the previous close is above it; the upper can only fall while that close is below it. Otherwise that boundary resets. Crossing the prior opposing boundary flips the trend.

2. Machine-learning models and completed probes

Probe Return Equation

$$g = s\frac{C - P}{A_{\text{entry}}}$$

g is the probe score; s is +1 for a long and -1 for a short. P is entry price, C the fifth-candle close and A entry the smoothed true range at entry. A one-ATR price rise gives g = +1 for the long and -1 for the short. Up to five probes per direction can overlap; each result enters learning after its fifth candle closes.

Three feedback routes propose settings. Rolling results propose wider, slower bands below a 38% win rate with a negative average score, and tighter, faster bands above 62% wins with a positive average. Small batches react to recent results, giving less weight to probes that moved far against the trade. An 8 by 8 grid stores outcomes across trend/range and volatility conditions. Its influence grows with supporting observations, up to 65% of the context/global blend.

The model blends proposals into a target and smooths the update. A larger multiplier requires a larger reversal to flip trend; a longer ATR length reacts more slowly to volatility changes. Learning changes signal timing. The account retains its separate position and loss limits.

Adaptive Update Equation

$$\theta_{t+1} = \theta_t + \alpha_t(\tilde{\theta}_t - \theta_t)$$

theta is the current setting; theta with a tilde is its rounded, bounded target; alpha is the update fraction. For a multiplier of 1.40, target 1.50 and alpha = 0.20, the next setting is 1.42. Deadbands ignore small changes; update caps and baseline reversion limit drift. Length remains 5-100 and multiplier 0.5-5.0.

3. Turn the adjusted indicator into a decision

BUY requires a stored fresh low, an upward trend flip, RSI(14) below 30 within 50 candles and at least 10 candles since the last signal. RSI compares smoothed gains and losses on a 0-100 scale. SELL uses a fresh-high reversal, downward flip and recent RSI above 70. Volume-surge filtering and live tick pressure are off; all decisions use confirmed candles.

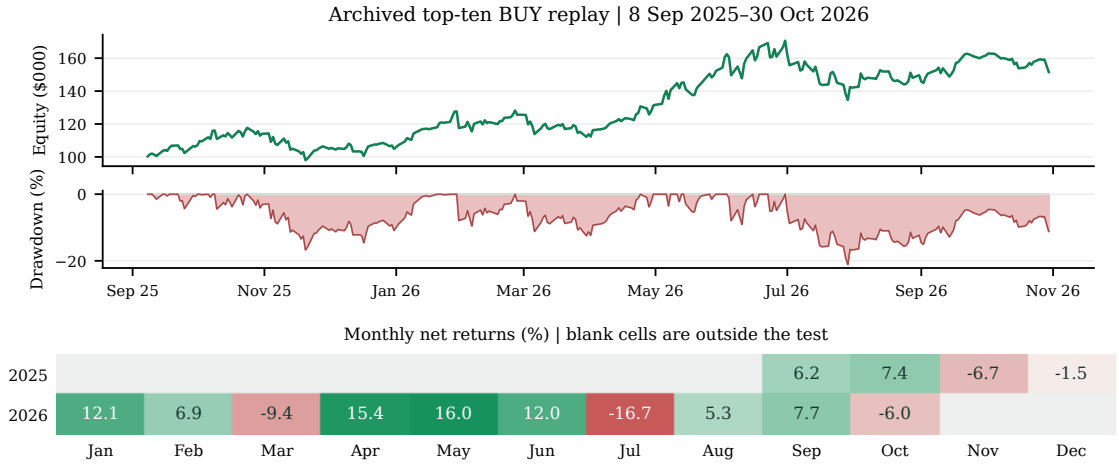
Raw inputs	Adaptive model	Investment decision
Completed 30-minute bars and matured five-candle probe results.	Weighted feedback updates band width and smoothing within their bounds.	A confirmed BUY passes liquidity and risk checks before the order.

Dai et al., *Reversals and the Returns to Liquidity Provision* (2023, NBER 30917), link higher volatility to faster, initially stronger reversals. This motivates adaptation; it does not validate our specific entry rule.

2 Archived signals and observed results

Our signal system emailed the ten strongest BUY candidates it found, ranked by score. We archived those dated messages and replayed the saved lists against real historical market prices, rather than recreating picks after seeing subsequent returns. The replay covers 8 September 2025-30 October 2026: 290 sessions, starting with $100,000.

Net total return	Annualized growth	Maximum drawdown	Final marked equity
+51.54%	+43.92%	-21.10%	$151,542.68



Data: archived top-ten BUY emails; Alpaca SIP daily bars, split-adjusted. Our replay and calculations. Costs: 5 basis points of fees plus 5 basis points of adverse slippage per side. September 2025 is a partial month.

The declared strategy exits when a SELL signal appears or a protective stop triggers. The earlier replay uses the first email each date, next-open fills, $10,000 per name and ten holdings maximum. Historical replay exits use ranking turnover, a 5% stop or 15% target. The last email, dated 14 October, is carried forward through 30 October; ten holdings remain marked open.

We reproduced all 1,254 simulated fills: 632 BUYs and 622 SELLs. Dates, tickers, shares, prices, fees and cash amounts match the saved ledger. The results describe the earlier email workflow; the current declaration uses the 30-minute signals and portfolio limits specified here.

Execution sensitivity

Replay case	Total return	Max. drawdown	Sessions
Zero modeled cost	64.45%	-19.47%	290
10 bps per side	51.54%	-21.10%	290
20 bps per side	39.11%	-22.81%	290
10 bps + one extra session	43.49%	-21.91%	289

Doubling modeled costs reduces return by 12.43 percentage points; delaying ranking eligibility reduces it by 8.05 points. The delayed case starts one session later; stop and target exits retain next-open execution. Allocation and exit thresholds are fixed across cases. These results show material sensitivity to execution costs and timing.

At 10 bps, 9 of 14 months are positive; closed-trade win rate is 46.62%. Daily Sharpe is 1.22 using 252 sessions per year and a zero risk-free rate. QQQ gains 19.39% over the same baseline dates, measured from the first open to the final close, without dividends or trading costs.

Research repository: github.com/V-Juneja/ml-supertrend
Replay code, dated rankings, result tables, charts and reproduction instructions: docs/BACKTEST.md.

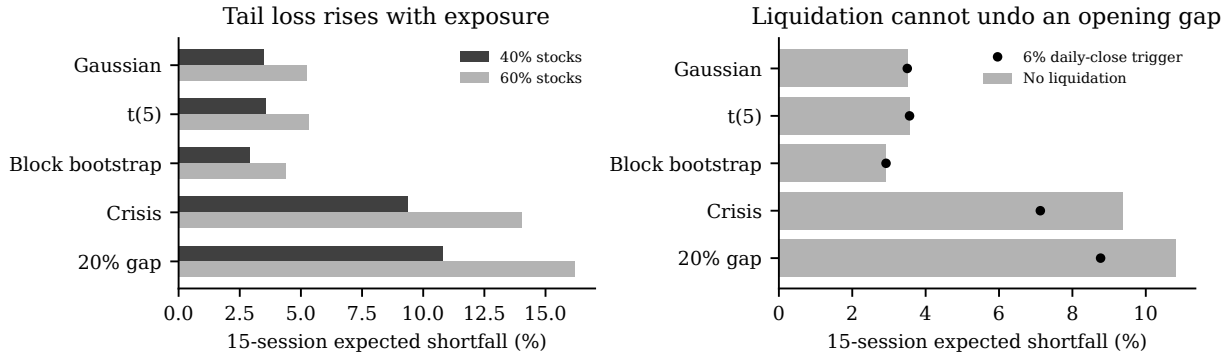
3 Monte Carlo results and decisions

From each stock's last regular-session close, we calculate log(close / prior close) on common dates without filling missing prices. The 59 daily returns, 2 July-25 September 2026, estimate six-stock covariance with 11.00% Ledoit-Wolf shrinkage toward a common-variance diagonal. Each model runs 50,000 fifteen-session paths with fixed quantities and zero cash interest.

Value at Risk and Expected Shortfall Equations

$$\mathrm{VaR}_{95} = Q_{0.95}(L), \qquad \mathrm{ES}_{95} = \mathrm{E}[L \mid L \geq Q_{0.95}(L)]$$

L is percentage account loss; Q with subscript 0.95 is its 95th percentile. E is an average, conditional on the event after the vertical line. VaR is exceeded in 5% of paths; ES is their average loss.



Data: Yahoo Finance daily closes, 2 Jul-25 Sep 2026; our calculations. The test divides 40% or 60% total stock exposure equally among the six calibration stocks.

Model at 40% exposure	95% ES	What the result means
Gaussian	3.50%	Baseline loss using fitted covariance and normal log-return shocks. This is the reference case for the other tests.
Student-t(5)	3.56%	Heavier-tailed log shocks give a similar 95% ES here. Similarity at this percentile does not settle the risk of rarer extremes.
Five-session block bootstrap	2.92%	Resampling joint five-session return blocks gives lower ES in this sample. It preserves co-movement but cannot add crashes absent from the short history.
Volatility / correlation crisis	9.36%	Raising volatility by 50% and correlation to 0.90 more than doubles baseline ES. This supports limiting common sector and theme exposure.
Common 20% opening gap	10.81%	The gap passes through intended stop prices. At the daily-close liquidation trigger, ES is still 8.77%, above the 6% trigger.

At 40% versus 60% exposure, Gaussian ES falls from 5.25% to 3.50%. Crisis ES is 2.67 times baseline; the gap-case guard cannot enforce a 6% loss ceiling. These are fixed-holding risk scenarios. Gaussian expected price drift is zero; the 59-return calibration is short. Seeds: 20261102-20261106.

Net Trade Expectancy Equation

$$e = pW - (1-p)D - c$$

e is the average net gain per trade as a percentage of entry capital. p is the fraction of trades winning before costs; W is their mean gross gain, D the positive magnitude of the mean gross loss, and c average round-trip costs. Positive expectancy means gains outweigh losses and costs across the sample. Individual trades can still lose. All returns use percentage units.

In the archived replay's 622 closed trades, p = 47.75%, W = 8.07%, D = 5.42% and c = 0.201%. Full-precision calculation gives e = +0.817%: $81.73 net per $10,000 position on average. It excludes ten open positions and describes historical trades, not a guaranteed next-trade gain. Calculation: GitHub backtest evidence.

4 Portfolio construction and risk

Risk limits address false reversals, gaps, liquidity and common sector exposure. The 40% gross-stock cap reduces modelled tail loss relative to 60% exposure. The limits below govern actual orders; they are declared design choices, not historically optimized thresholds.

Control	Execution rule and purpose
Exposure	Long-only, no borrowing; at most six names, 10% of equity per name and 40% gross stocks. Unused capacity stays uninvested.
Loss budget	Initial stop two 30-minute ATR(24) below fill; planned risk at most 0.5% of equity per name and 2% across the account. Never widen a stop.
Concentration	At most 20% in one sector and 25% in the combined semiconductor / technology-infrastructure cluster. Correlated names consume shared capacity.
Liquidity	Price at least $5; trailing 20-session average dollar volume at least $20 million. Reject stale, halted or incomplete quotes.
Account losses	At 2% session loss, pause entries until the next session. At 6% from peak marked equity, liquidate at available prices and halt for the competition.
Execution integrity	One account, one executor; no duplicate BUY entries. Log signal and parameter state, order, fill, fees, stop and exit reason. Close by the required final cutoff.

Position Sizing Equation

$$d_i = 2\,\mathrm{ATR}_{24,i}, \qquad q_i = \left\lfloor \min\left(\frac{0.10E}{P_i}, \frac{0.005E}{d_i}\right) \right\rfloor$$

For stock i, E is marked account equity, P its entry price, ATR with subscript 24 its 24-candle smoothed true range, d the dollar stop distance and q whole shares. Floor brackets round down. The smaller value/risk cap governs; remaining exposure, sector capacity, aggregate risk and costs can reduce the order.

At DELL's recorded 25 September close of $563.17, a $10,000 value cap permits 17 shares, or $9,573.89. In a $100,000 account, that order exhausts the $500 risk allowance at a $29.41 stop distance; a wider two-ATR stop requires fewer shares. This is a sizing calculation, not an observed trade. Portfolio capacity can reduce it further.

Stop-loss overshoot and common shocks

Stop execution stress	Loss / planned stop loss	Account loss at 2% risk budget
Fills at intended stop	1.0 x	2%
Loss exceeds stop distance by 50%	1.5 x	3%
Loss doubles the intended distance	2.0 x	4%
Gap produces three times planned loss	3.0 x	6%

Calculation: aggregate planned loss multiplied by the realized loss/stop-distance ratio. This is a deterministic exposure test, not a forecast. Gaps can exceed all ratios shown; neither a stop nor the 6% liquidation trigger guarantees a maximum loss.

A separate 72-case stress grid crosses 15/126-session horizons, 20/40/60% exposure, 25/45/65% annual volatility and correlations 0/0.35/0.65/0.90, with 100,000 draws per case. Daily Gaussian, t(5), common-gap and liquidation tests use 40,000 paths per model. At 40% exposure, a simultaneous 10% stock fall costs 4%; a 20% fall costs 8%. Sector and shared-theme caps address positions exposed to the same economic driver.

Final declared strategy

Trade confirmed 30-minute ML Supertrend reversals, long-only. Only matured five-candle probes update settings within the declared bounds. Size qualifying BUYs by the position sizing equation, subject to six names, 10% per name, 40% stock exposure and 2% aggregate planned risk. Exit on SELL or stop; pause entries after a 2% session loss and liquidate and halt at 6% account drawdown. Freeze the rules and log accepted and rejected signals.